BT5110 Tutorial 4

Normalization

Your company, Apasaja Private Limited, is commissioned by Toko Kopi Luwak to design a relational schema for the management of their coffee beans, drinks, and cafes.

A coffee bean is identified by its unique brand name or a combination of its cultivar and region (since the same cultivar can be grown in different regions). For instance, we may have a coffee bean named “The Waterfall” which comes from a Tabi cultivar grown in Colombia.

A drink can be made using a particular coffee bean. The name of the drink is only unique for a particular coffee bean. This means we can have an “Espresso” made with “The Waterfall” or “La Bella” (which is a Pacamara cultivar grown in Guatemala). The price of the drink is also recorded.

A branch identified by its branch name may sell the drink. A drink may be sold by zero or more branches, and a branch may sell zero or more drinks. The address of the branch is also recorded. For each drink sold by a branch, we record the quantity sold to see which branch is the most profitable.

An abstract schema for this application is given below.

$$R = \{A, B, C, D, E, F, G, H\}$$

$$\Sigma = \{ \{A\} \rightarrow \{C, E\}, \{A, B\} \rightarrow \{D\}, \{F\} \rightarrow \{H\}, \{C, E\} \rightarrow \{A\}, \{B, C, E\} \rightarrow \{D\}, \{A, B, F\} \rightarrow \{D, G\}, \{B, C, E, F\} \rightarrow \{G\} \}$$

1. Normal Form

(a) Is R with Σ in 3NF?

(b) Is R with Σ in BCNF?

2. Normalization

(a) Decompose R with Σ into a lossless-join 3NF decomposition using the algorithm from the lecture.

(b) Is the result dependency preserving?

(c) Is the result in BCNF?